12.463

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Question

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 4 \\ -1 & -1 & -2 \end{pmatrix}$$

whose eigenvalues are $1, -1$ and 3. Then Trace of $\left(\mathbf{A}^3 - 3\mathbf{A}^2\right)$ is ______.

Theoretical Solution

The eigenvalues of $\mathbf{A}$ are given as $\lambda_1 = 1$, $\lambda_2 = -1$, and $\lambda_3 = 3$.

We need to find the trace of the matrix $\mathbf{B} = \mathbf{A}^3 - 3\mathbf{A}^2$. This is a polynomial of the matrix $\mathbf{A}$, where the polynomial is $p(x) = x^3 - 3x^2$.

If a matrix $\mathbf{A}$ has eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, then for any polynomial $p(x)$, the matrix $p(\mathbf{A})$ will have eigenvalues $p(\lambda_1), p(\lambda_2), \ldots, p(\lambda_n)$.

Theoretical Solution

Using this property, the eigenvalues of $\mathbf{B}$ (let's call them μ_i) are $p(\lambda_i)$.

- For $\lambda_1 = 1$:
$$\mu_1 = (1)^3 - 3(1)^2 = 1 - 3 = -2$$
- For $\lambda_2 = -1$:
$$\mu_2 = (-1)^3 - 3(-1)^2 = -1 - 3(1) = -1 - 3 = -4$$
- For $\lambda_3 = 3$:
$$\mu_3 = (3)^3 - 3(3)^2 = 27 - 3(9) = 27 - 27 = 0$$

So, the eigenvalues of the matrix $(\mathbf{A}^3 - 3\mathbf{A}^2)$ are -2, -4, and 0.

Theoretical Solution

The trace of a matrix is the sum of its eigenvalues.

$$\begin{aligned}\text{Trace}(\mathbf{A}^3 - 3\mathbf{A}^2) &= \mu_1 + \mu_2 + \mu_3 \\ &= (-2) + (-4) + 0 \\ &= -6\end{aligned}$$

The Trace of $(\mathbf{A}^3 - 3\mathbf{A}^2)$ is -6.